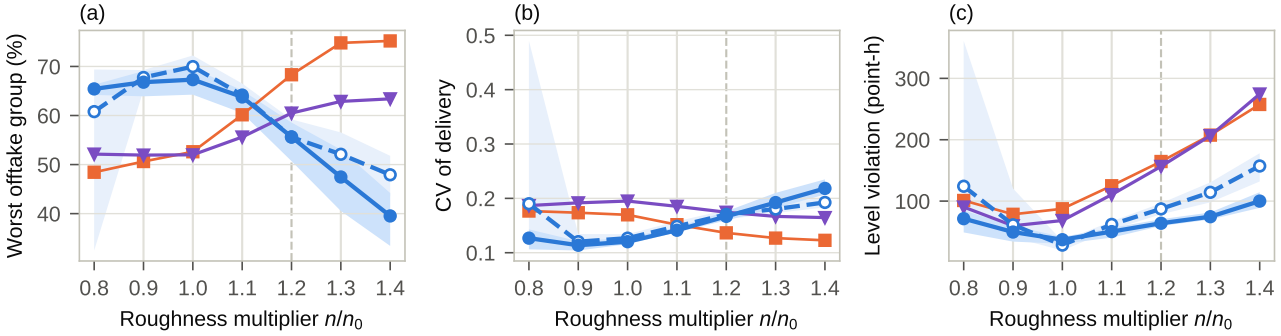
(a)
Worst offtake group (%)
70
60
50
40
0.8
0.9
1.0
1.1
1.2
1.3
1.4
Roughness multiplier n/n_0
(b)
CV of delivery
0.5
0.4
0.3
0.2
0.1
Roughness multiplier n/n_0
(c)
Level violation (point-h)
300
200
100
Roughness multiplier n/n_0
Fixed rule + FF/P (13 gates)
Staged rule, randomized training, mean of 3 seeds
Neural, nominal training (n/n_0 = 1), 140 gen.
Neural, randomized roughness (n/n_0 = 0.9, 1.0, 1.1, 1.3), 140 gen.
Test value (not used in training)
Randomized training values